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 Oefeningenreeks 6A nr. 9.1

$$\text{RR: } \begin{cases} x_{n-1} + x_n + x_{n+1} = 51(1) \\ x_{n-1} \cdot x_n \cdot x_{n+1} = 4641(2) \end{cases}$$

$$x_{n+1} = x_n + v \text{ en } x_{n-1} = x_n - v$$

$$(1) : x_{n-1} + x_n + x_{n+1} = 51$$

$$\Rightarrow 3x_n = 51$$

$$\Rightarrow x_n = 17$$

$$(2) : (x_n - v)x_n(x_n + v) = 4641$$

$$\Rightarrow x_n(x_n^2 - v^2) = 4641$$

$$\Rightarrow 17(17^2 - v^2) = 4641$$

$$\Rightarrow v^2 = 16$$

$$\Rightarrow v = \pm 4$$

$$\Rightarrow (x_n)_n = (13, 17, 21) \text{ of } (x_n)_n = (21, 17, 13)$$

References